



WINTRIX ACADEMY PRESENTS

DCET Previous Year Question Paper & Solutions

2023

DIPLOMA COMMON ENTRANCE TEST

GUIDANCE • STRATEGY • SUCCESS

100 Questions • Detailed Solutions • Subject-Wise Organisation

FIRST EDITION — 2026



DCET 2023 Previous Year Question Paper & Solutions

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DCET All-Karnataka Rank 48 | RVCE CSE | Placed at Red Hat

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Disclaimer: This book contains previous year question papers from the DCET examination conducted by the Karnataka Examination Authority (KEA). The solutions and explanations provided are prepared by Wintrix Academy for educational purposes. While every effort has been made to ensure accuracy, readers are advised to verify critical information independently.

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About Wintrix Academy

Wintrix Academy is a premier DCET mentorship platform founded with a singular mission: to empower Diploma students across Karnataka with the guidance, strategy, and resources they need to secure top ranks in the Diploma Common Entrance Test and gain admission to the state's most prestigious engineering institutions.

Built on the foundation of real experience and proven results, Wintrix Academy bridges the gap between aspiration and achievement through structured mentorship, comprehensive study materials, and a supportive learning ecosystem.



Mohammed Adnan

FOUNDER & CHIEF MENTOR

Mohammed Adnan completed his SSLC at CKS with 91.52% and pursued a Diploma in Electrical & Electronics Engineering, graduating with a CGPA of 9.8. He secured All-Karnataka DCET Rank 48 and 2nd Rank in Hassan District, earning admission to R.V. College of Engineering (RVCE) for Computer Science Engineering, where he currently maintains a CGPA of 9.19. His dedication to excellence has led to a placement at Red Hat, a renowned US-based technology company.

48

DCET RANK

9.8

DIPLOMA
CGPA

9.19

RVCE CGPA

50+

STUDENTS



Jnanesh Gowda

DIRECTOR & BRAND ARCHITECT

Turning confusion into clarity and effort into results — that's what drives Wintrix forward. Jnanesh brings new-age learning strategies, purpose-driven outreach, structured content design, and results-engineered systems to ensure every student's success is built by design, not left to chance.

OUR PROMISE: At Wintrix Academy, success isn't left to chance — it's engineered from day one. We provide live classes, recorded lectures, mock tests, rankings, 24/7 support, and a thriving community of DCET aspirants.

Preface

Dear DCET Aspirant,

Preparing for the Diploma Common Entrance Test is a journey that demands clarity, consistency, and the right resources. Having walked this very path and secured All-Karnataka Rank 48, I understand the challenges, the uncertainties, and the relentless effort it takes to achieve a top rank.

This book has been meticulously crafted to serve as your companion in that journey. It contains the complete DCET 2023 Question Paper along with detailed, step-by-step solutions for all 100 questions. Every solution has been prepared with the goal of not just giving you the right answer, but helping you understand the reasoning behind it.

The questions are organised subject-wise across five chapters — Engineering Mathematics, Statistics & Analytics, IT Skills, Fundamentals of Electrical & Electronics Engineering, and Project Management Skills — each carrying 20 marks. This structure mirrors the actual exam pattern and allows you to focus your preparation systematically.

Practising previous year papers is one of the most effective strategies for DCET preparation. It helps you understand the exam pattern, identify frequently tested topics, manage your time effectively, and build the confidence you need to perform at your best on exam day.

I encourage you to attempt each question on your own before looking at the solution. Mark the questions you find challenging and revisit them until you've mastered the concept. Remember, every question you solve brings you one step closer to your dream college.

Wishing you all the very best for your DCET journey. May your hard work translate into the rank you deserve.

Mohammed Adnan

Founder, Wintrix Academy
DCET Rank 48 | RVCE CSE

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100 QUESTIONS • 5 SUBJECTS • 20 MARKS EACH • 100 MARKS TOTAL

Introduction to D-CET

DCET — Diploma Common Entrance Test — is conducted by the Karnataka Examination Authority (KEA). The entrance exam is a requirement for enrolment in the second-year undergraduate engineering programmes offered through the lateral entry programme. In Karnataka, these programmes are offered at both day and evening colleges.

D-CET Exam Pattern

Duration	3 Hours
Maximum Marks	100
Type of Questions	Multiple Choice (MCQ)
Total Questions	100
Subjects Covered	5 Subjects

D-CET Tentative Important Dates

Application form starts	2nd Week of April
Last date for application submission	4th Week of April
D-CET exam date	Month of May
Counselling date	Month of June

Subject-wise Mark Distribution

1. Engineering Mathematics	20 Marks
2. Statistics and Analytics	20 Marks
3. IT Skills	20 Marks
4. Fundamentals of Electrical & Electronics Engineering	20 Marks
5. Project Management Skills	20 Marks
Total	100 Marks

01

CHAPTER ONE

Engineering Mathematics

20 Marks

Questions 1 – 20

Q.01

MATRICES

In a square matrix, if the elements above the principal diagonal are zero, then it is called

- a) Identity matrix **b) Lower triangular matrix**
- c) Upper triangular matrix d) Diagonal matrix

✓ ANSWER: (B)

In a square matrix, if all the elements above the principal diagonal are zero, it is called a **Lower Triangular Matrix**.

Q.02

DETERMINANTS

The value of x if the matrix $[x-1, 2; 2, 4]$ is singular, is

- a) 3 b) 8
- c) -2 **d) 2**

✓ ANSWER: (D)

Given matrix $A = [x-1, 2; 2, 4]$ is singular $\Rightarrow |A| = 0$

$$(x-1) \cdot 4 - 2 \cdot 2 = 0$$

$$4x - 4 - 4 = 0$$

$$4x = 8$$

$$\therefore x = 2$$

Q.03

MATRIX INVERSE

The inverse of the matrix $A = [-1, 0; 5, 7]$ is

a) $-1/7 [7, 0; -5, -1]$

b) $1/7 [7, 0; 5, 1]$

c) $-1/7 [-7, 0; 5, -1]$

d) $1/7 [7, 0; 5, 1]$

✓ ANSWER: (A)

Given $A = [-1, 0; 5, 7]$

$$|A| = (-1)(7) - (0)(5) = -7$$

$$A^{-1} = (1/|A|) \cdot \text{adj}(A) = -1/7 \cdot [7, 0; -5, -1]$$

Q.04

EIGENVALUES

The eigenvalue of the matrix $A = [3, 0; 1, 3]$ is

a) 2, 2

b) -3, -3

c) 3, 3

d) -3, 3

✓ ANSWER: (C)

The characteristic equation of A is $|A - \lambda I| = 0$

$$|3-\lambda, 0; 1, 3-\lambda| = 0$$

$$(3-\lambda)(3-\lambda) - 0 = 0$$

$$(3-\lambda)^2 = 0$$

$$\therefore \lambda = 3, \lambda = 3$$

Q.05

STRAIGHT LINES

The two lines $ax + by = c$ and $a'x + b'y = c'$ are perpendicular if

a) $ab' = ba'$

b) $aa' + bb' = 0$

c) $ab + a'b = 0$

d) $ab + ba = 0$

✓ ANSWER: (B)

For line $ax + by = c$, slope $m_1 = -a/b$. For line $a'x + b'y = c'$, slope $m_2 = -a'/b'$.

Lines are perpendicular when $m_1 \times m_2 = -1$

$$(-a/b)(-a'/b') = -1$$

$$aa'/bb' = -1$$

$$\therefore aa' + bb' = 0$$

Q.06

STRAIGHT LINES

The y-intercept of any line passing through the origin is

a) 0

b) 1

c) -1

d) 2

✓ ANSWER: (A)

The y-intercept of any line passing through the origin is **zero**, since the line crosses the y-axis at the point (0, 0).

Q.07

STRAIGHT LINES

Slope intercept form of straight line is

a) $y + mx - c = 0$

b) $x = my + c$

c) $y = x + m$

d) $y = mx + c$

✓ ANSWER: (D)

The slope-intercept form of a straight line is **$y = mx + c$** , where m is the slope and c is the y-intercept.

Q.08

STRAIGHT LINES

The tangent of the angle of intersection between two lines with slopes m_1 and m_2 is

- a) $(m_1 + m_2) / (1 - m_1 m_2)$ b) $(m_1 + m_2) / (m_1 - m_2)$
c) $(m_1 - m_2) / (1 + m_1 m_2)$ d) None of these

✓ ANSWER: (C)

The tangent of the angle of intersection between two lines with slopes m_1 and m_2 is

$$\tan \theta = (m_1 - m_2) / (1 + m_1 m_2)$$

Q.09

TRIGONOMETRY

If the ladder is inclined to the wall at an angle of 135° , then the inclination in radians is

- a) $\pi/4$ b) $\pi/2$
c) $3\pi/4$ d) $2\pi/3$

✓ ANSWER: (C)

Given angle $\theta = 135^\circ$

$$\text{Angle in radians} = (\pi/180) \times 135 = 135\pi/180 = 3\pi/4 \text{ radians}$$

Q.10

TRIGONOMETRY

The value of $\sin 60^\circ \cos 30^\circ - \cos 60^\circ \sin 30^\circ$ is

- a) 0 b) $1/2$
c) $\sqrt{3}/2$ d) 1

✓ ANSWER: (B)

Using the formula $\sin(A - B) = \sin A \cos B - \cos A \sin B$:

$$\sin 60^\circ \cos 30^\circ - \cos 60^\circ \sin 30^\circ = \sin(60^\circ - 30^\circ) = \sin 30^\circ = 1/2$$

Q.11

The simplified value of $(\sin 3A + \sin A) / (\sin 3A - \sin A)$ is

- a) $\cot A \tan 5A$ b) $\tan A \cot 2A$
c) $\tan 2A \cot A$ d) $\tan 3A \cot 2A$

✓ ANSWER: (C)

Using transformation formulae:

$$\sin C + \sin D = 2 \sin((C+D)/2) \cos((C-D)/2)$$

$$\sin C - \sin D = 2 \cos((C+D)/2) \sin((C-D)/2)$$

$$\begin{aligned} & (\sin 3A + \sin A) / (\sin 3A - \sin A) \\ &= [2 \sin 2A \cos A] / [2 \cos 2A \sin A] \\ &= \tan 2A \cdot \cot A \end{aligned}$$

Q.12

If $y = \log x + \sec 2x$, then dy/dx is

- a) $1/x + \sec 2x \tan x$
- b) $1/x + 2 \sec 2x \tan 2x$
- c) $1/x^2 + \sec x \tan x$
- d) $1/x^2 + \sec 3x \tan 2x$

✓ ANSWER: (B)

$$y = \log x + \sec 2x$$

$$dy/dx = 1/x + 2 \sec 2x \cdot \tan 2x$$

The derivative of the function $(1 + x) / (1 - x)$ is

- a) $1/(1-x)^2$ b) $-2/(1-x)^2$
 c) $x/(1-x)^2$ d) $2/(1-x)^2$

✓ ANSWER: (D)

Using quotient rule: $y = (ax+b)/(cx+d)$, then $dy/dx = (ad-bc)/(cx+d)^2$

$$\begin{aligned} y &= (1+x)/(1-x) \\ dy/dx &= [(1)(-1) - (1)(1)] \cdot (-1) / (1-x)^2 \\ &= 2/(1-x)^2 \end{aligned}$$

Find the second order derivative of $y = e^{2x} - e^x$

- a) $4e^{2x} - e^x$ b) $4e^{2x} + e^x$
 c) $-4e^{2x} - e^x$ d) $4e^x + e^x$

✓ ANSWER: (A)

$$\begin{aligned} y &= e^{2x} - e^x \\ dy/dx &= 2e^{2x} - e^x \\ d^2y/dx^2 &= 4e^{2x} - e^x \end{aligned}$$

The equation of the tangent to the curve $y = 2x^2 + x$ at $(1, 2)$ is

a) $5x - y - 3 = 0$

b) $5x + y + 3 = 0$

c) $5x + y - 6 = 0$

d) $5x + y + 6 = 0$

✓ ANSWER: (A)

$$y = 2x^2 + x$$

$$dy/dx = 4x + 1$$

$$\text{At } (1, 2): m = dy/dx|_{(1)} = 4(1) + 1 = 5$$

$$\text{Equation of tangent: } y - y_1 = m(x - x_1)$$

$$y - 2 = 5(x - 1)$$

$$y - 2 = 5x - 5$$

$$\therefore 5x - y - 3 = 0$$

The value of $\int (\sec x \tan x + \sec^2 x) dx$ is

a) $\sec 2x + \tan x + c$

b) $\sec x + \operatorname{cosec} x + c$

c) $\sec x + \tan x + c$

d) $\sec 2x - \operatorname{cosec}^2 x + c$

✓ ANSWER: (C)

$$\int (\sec x \tan x + \sec^2 x) dx = \sec x + \tan x + c$$

The value of $\int (2x^3 + 3x^2 + 2x)^{10} (3x^2 + 3x + 1) dx$ is

- a) $\frac{1}{2} \cdot (6x^3 + 3x^2 + 2x)^{25} + c$ **b) $\frac{1}{22} \cdot (2x^3 + 3x^2 + 2x)^{11} + c$**
- c) $\frac{1}{12} \cdot (2x^3 + 3x^2 - 2x)^{12} + c$ d) $\frac{1}{12} \cdot (2x^3 + 3x^2 - 2x)^{12} + c$

✓ ANSWER: (B)

Using the formula $\int [f(x)]^n f'(x) dx = [f(x)]^{n+1}/(n+1) + c$

$$\begin{aligned} \text{Let } f(x) &= 2x^3 + 3x^2 + 2x \\ f'(x) &= 6x^2 + 6x + 2 = 2(3x^2 + 3x + 1) \end{aligned}$$

$$\begin{aligned} \int (2x^3 + 3x^2 + 2x)^{10} (3x^2 + 3x + 1) dx \\ &= \frac{1}{2} \int (2x^3 + 3x^2 + 2x)^{10} (6x^2 + 6x + 2) dx \\ &= \frac{1}{2} \cdot (2x^3 + 3x^2 + 2x)^{11} / 11 + c \\ &= \frac{1}{22} \cdot (2x^3 + 3x^2 + 2x)^{11} + c \end{aligned}$$

The value of $\int_0^{\pi/4} \tan^2 x dx$ is

- a) $1 + \pi/4$ b) $1 + 4/\pi$
- c) $1 - \pi/4$** d) $1 - 4/\pi$

✓ ANSWER: (C)

$$\begin{aligned} \int_0^{\pi/4} \tan^2 x dx &= \int_0^{\pi/4} (\sec^2 x - 1) dx \\ &= [\tan x - x]_0^{\pi/4} \\ &= (\tan \pi/4 - \pi/4) - (\tan 0 - 0) \\ &= 1 - \pi/4 \end{aligned}$$

The area bounded by $y = \sin x$ and x-axis from $x = 0$ to $x = \pi$ is

a) 2

b) -2

c) 3

d) 1

✓ ANSWER: (A)

$$\begin{aligned}\text{Area} &= \int_0^\pi \sin x \, dx = [-\cos x]_0^\pi \\ &= -\cos \pi - (-\cos 0) = -(-1) + 1 = 1 + 1 = 2\end{aligned}$$

The value of $\tan 45^\circ \cdot \cot 225^\circ + \tan^2 60^\circ$ is

a) -4

b) 4

c) 2

d) 3

✓ ANSWER: (B)

$$\begin{aligned}\tan 45^\circ \cdot \cot 225^\circ + \tan^2 60^\circ \\ &= 1 \times \cot(180^\circ + 45^\circ) + (\sqrt{3})^2 \\ &= 1 \times \cot 45^\circ + 3 \\ &= (1 \times 1) + 3 = 1 + 3 = 4\end{aligned}$$

02

CHAPTER TWO

Statistics & Analytics

20 Marks

Questions 21 – 40

Q.21

DATA TYPES

_____ is an example of quantitative data.

- a) Volume
- b) Words
- c) Symbols
- d) Colour

✓ ANSWER: (A)

The data which can be numerically measurable is called **quantitative data**. "Volume" is measured in numbers, hence it is quantitative data.

Q.22

DATA CLEANING

Data cleaning is the process of

- a) removing viruses
- b) correctly formatting data
- c) removing duplicate data
- d) properly formatting data

GRACE MARKS

Data cleaning is the process of fixing or removing incorrect, corrupted, incorrectly formatted, duplicate or incomplete data within a dataset.

Q.23

DATA COLLECTION

_____ is not a data collection tool.

- a) Word
- b) Focus Group Discussion
- c) Survey
- d) Questionnaires

✓ ANSWER: (A)

Data collection tools include focus group discussion, survey method, questionnaires, observation, and interview method. **Word** is not a data collection tool.

The graph of cumulative frequency is called

- a) Frequency polygon
- b) Histogram
- c) Cumulative frequency polygon
- d) Frequency histogram

GRACE MARKS

The cumulative frequencies can be plotted on a **cumulative frequency polygon** or cumulative frequency curve (Ogive curve).

To calculate percentage frequency, we use _____ formula.

- a) $P.f. = (f \times n) + 100$
- b) $P.f. = (f \div n) \times 100$
- c) $P.f. = (100) \div (f \times n)$
- d) $P.f. = (100) \times (f \div n)$

✓ ANSWER: (B) & (D)

Percentage frequency = $(f/n) \times 100$ where f is the frequency and n is the total number of observations.

If $X_1, X_2, X_3, \dots, X_n$ are the observations of a given data, then the mean of the observations will be:

- a) Total number of observations / Sum of observations
- b) Sum of observations + Total number of observations
- c) Sum of observations / Total number of observations
- d) Total number of observations – Sum of observations

✓ ANSWER: (C)

Mean = $\text{Sum of observations} / \text{Total number of observations}$

Q.27**CLASS INTERVALS**

The end points of a class interval are the _____ and _____ values that a variable can take.

- a) Lowest and Highest b) Minimum and Maximum
c) Numeral and Average d) Mean and Mode

✓ ANSWER: (A)

If [10–20] is a class interval, the end points are 10 and 20, which are the **lowest and highest** values of the variable.

Q.28**DATA INTERPRETATION**

The graph below shows the participation of students in outdoor games. In which years did the girls participate more than the boys?

DATA GIVEN: 2020: Boys=225, Girls=125 | 2021: Boys=125, Girls=225 | 2022: Boys=50, Girls=125 | 2023: Boys=225, Girls=150

- a) 2020, 2023 b) 2021, 2022
c) 2022, 2023 d) 2020, 2021

✓ ANSWER: (B)

From the data: In 2021 (Girls=225 > Boys=125) and 2022 (Girls=125 > Boys=50), girls participated more than boys.

Q.29**DATA INTERPRETATION**

In which two years did an equal number of boys participate?

- a) 2020, 2021 b) 2020, 2022
c) 2020, 2023 d) 2021, 2022

✓ ANSWER: (C)

In 2020 boys participation = 225, and in 2023 boys participation = 225. Hence equal.

Q.30

EXCEL

To find third quartile in Excel, we use _____ formula.

- a) =QUARTER (3, Range) b) =QUARTILE (3, Range)
- c) =QUARTER (Range, 3) d) =QUARTILE (Range, 3)

✓ ANSWER: (D)

The formula for finding the 3rd quartile in Excel is =QUARTILE (Range , 3)

Q.31

PERCENTILE

The percentile divides a series into _____ equal parts.

- a) fifty b) twenty
- c) ten d) hundred

✓ ANSWER: (D)

Data which divides into 100 equal parts is called percentile. It is denoted by P.

Q.32

QUARTILES

The value of first quartile is 23 and interquartile range of set of data is 20. The value of third quartile is

- a) 23 b) 33
- c) 42 d) 43

✓ ANSWER: (D)

$$Q_1 = 23, \text{ IQR} = Q_3 - Q_1 = 20$$

$$Q_3 - 23 = 20$$

$$Q_3 = 20 + 23 = 43$$

Q.33

MEAN

The algebraic sum of the deviations of a frequency distribution from its mean is always

- a) a non-zero number
- b) zero
- c) less than zero
- d) greater than zero

✓ ANSWER: (B)

The property of arithmetic mean states that the algebraic sum of the deviations from the mean is always equal to **zero**. i.e., $\sum(X - \bar{X}) = 0$

Q.34

EXCEL

The Excel formula for "Mean" is

- a) =MEDIAN (array of numbers)
- b) =AVERAGE (array of numbers)
- c) =MEAN (array of numbers)
- d) =MODE (array of numbers)

✓ ANSWER: (B)

The Excel formula for "mean" is **=AVERAGE(array of numbers)**

Q.35

PYTHON

What is output syntax in Python?

- a) Print(" ")
- b) PRINT(" ")
- c) print(" ")
- d) Printf (" ")

✓ ANSWER: (C)

In Python, **print(" ")** statement is used for output. Python is case-sensitive.

Q.36

PYTHON

"str" is a _____

- a) Text Type
- b) Numeric Type
- c) Binary Type
- d) Sequence Type

✓ ANSWER: (A)

str is a Text Type in Python.

Q.37

PYTHON

In Python, _____ standard data types are commonly used.

- a) three
- b) five
- c) ten
- d) four

GRACE MARKS

In Python, standard data types are: Numeric, Dictionary, Boolean, Set, and Sequence types.

Q.38

PYTHON

The result of Python program gets displayed in _____

- a) IDLE Shell 3.9.1 window
- b) IDLE Shell 3.1.9 window
- c) ILDE Shell 3.9.1 window
- d) IELD Shell 3.9.1 window

✓ ANSWER: (A)

IDLE 3.9.1 is the Integrated Development and Learning Environment for Python.

Which one of the following quotations in Python does not accept the quotes to denote strings?

a) ('')

b) (" ")

c) ()

d) ("" "")

✓ ANSWER: (C)

In Python, it accepts single quotes, double quotes, and triple quotes. Parentheses () without quotes do not denote strings.

In Python, _____ is used to end the physical line or ignore the comment.

a) **

b) #

c) &

d) \

✓ ANSWER: (B)

In Python, the # character is used to start a comment. The comment continues after # until the end of the line.

03

CHAPTER THREE

IT Skills

20 Marks

Questions 41 – 60

Q.41

CYBER SECURITY

Which of the following is not a cyber crime?

- a) Cryptography
- b) Denial of Service
- c) Man-in-the-middle attack
- d) Phishing

✓ ANSWER: (A)

Cryptography is the process of hiding or coding information so that only the intended recipient can read it. It is a security measure, not a cyber crime.

Q.42

CYBER SECURITY

DoS is abbreviated as

- a) Denial of Service
- b) Distribution of Server
- c) Distribution of Service
- d) Denial of Server

✓ ANSWER: (A)

DoS stands for **Denial-of-Service**.

Q.43

CYBER SECURITY

_____ protects interconnected systems including hardware, software and programs and data from cyber attacks.

- a) Cyber Security
- b) Computer Security
- c) Resource Security
- d) Hardware Security

✓ ANSWER: (A)

Cyber Security is the protection of interconnected systems including hardware, software, programs, and data from cyber attacks.

Basic functionality of the network device firewall is:

- a) scans mobile applications
- b) monitoring database
- c) privatizes the computers
- d) monitoring incoming and outgoing networks

✓ ANSWER: (D)

A **firewall** is a network security device that monitors incoming and outgoing network traffic and decides whether to allow or block specific traffic based on a defined set of security rules.

An algorithm represented in the form of programming language is called:

- a) Flowchart
- b) Pseudocode
- c) Program
- d) Instruction

✓ ANSWER: (C)

An algorithm represented in the form of a programming language is called a **program**.

The _____ symbol is used when the flowchart is starting or ending.

- a) Connector/Arrow
- b) Terminal box/Rounded rectangle
- c) Input/Output
- d) Process

✓ ANSWER: (B)

The **Terminator Symbol** (rounded rectangle) represents the start and end points of a flowchart.

MIT App Inventor allows user to

- a) Create web application
- b) Build Android application**
- c) Create System Software
- d) Develop Operating System

✓ ANSWER: (B)

MIT App Inventor allows users to build fully functional apps for smartphones and tablets.

What is the function of the "when green flag clicked" command block?

- a) Points Sprite in the specified direction
- b) If condition is true, runs the blocks inside
- c) Runs the script**
- d) Stops the execution of script

✓ ANSWER: (C)

The "when green flag clicked" block triggers the execution of the script or program attached to it.

The correct sequence of HTML tags for starting a webpage is

- a) Head, Title, Html, Body
- b) Html, Head, Title, Body**
- c) Html, Body, Title, Head
- d) Html, Title, Head, Body

✓ ANSWER: (B)

The correct sequence of HTML tags to start a webpage is `<html>, <head>, <title>, <body>`

Web server:

- a)** is a computer system that delivers web pages
- b) is delivery news
- c) provides options for those seeking real-time discussions
- d) prints documents

✓ ANSWER: (A)

A **web server** stores and delivers the content for a website.

Which of the following is used to style the appearance of web pages?

- a) HTML
- b) JavaScript
- c) PHP
- d) CSS**

✓ ANSWER: (D)

CSS is used to define styles for web pages, including design, layout, and variations in display for different devices and screen sizes.

Which of the following is an example of a web browser?

- a) Google
- b) Firefox**
- c) Apache
- d) MySQL

✓ ANSWER: (B)

Firefox is an example of a web browser.

Which of the following is an open source and free workflow management software?

- a) Trello
- b) MS Excel
- c) Windows
- d) Linux

✓ ANSWER: (A)

Trello is the visual tool that empowers teams to manage any type of project, workflow, or task tracking.

ERP package will handle _____ business functionality/functionalities.

- a) One
- b) Two
- c) Three
- d) Multiple/all

✓ ANSWER: (D)

ERP systems support all aspects of modules and provide transparency into the complete business process.

_____ is a visual diagram of a company that describes what employees do, whom they report to and how decisions are made across the business.

- a) Physical Structure
- b) Organizational Structure
- c) Logical Structure
- d) Hybrid Structure

✓ ANSWER: (B)

Organisational structure describes the entire structure of an organisation and work structure of employees.

_____ is a methodology used in system analysis to identify, clarify, and organize system requirements.

- a) Workflow
- b) Use case
- c) Algorithm
- d) Software

✓ ANSWER: (A)

Workflow helps in identifying, clarifying, and organising requirements of a system.

Which of the following is not an application of IoT?

- a) Web browser
- b) Smart home
- c) Smart city
- d) Self-driven cars

✓ ANSWER: (A)

Smart home, smart city, and self-driving cars are applications of IoT, but a **web browser** is not an IoT application.

Which of the following is not a cloud service option?

- a) VaaS
- b) IaaS
- c) PaaS
- d) SaaS

✓ ANSWER: (A)

PaaS, SaaS, and IaaS are the three cloud services. **VaaS** (Voice as a Service) is not a standard cloud service model.

CLOUD

d) 5

Cloud computing offers **3 types** of services: PaaS, SaaS, and IaaS.

CLOUD

d) Private

Hybrid cloud refers to a mixed computing, storage, and services environment made up of on-premises infrastructure, private cloud services, and a public cloud.

04

CHAPTER FOUR

Fundamentals of Electrical & Electronics Engineering

20 Marks

Questions 61 – 80

Q.61

ELECTRICAL BASICS

Unit of electrical power is

- a) Volt
- b) Watt
- c) Watt-hour
- d) Ampere-hour

✓ ANSWER: (B)

A **Watt** is the unit of electrical power equal to one ampere under the pressure of one volt. 1 W is the power consumed by a device that carries 1 A of current when operated at a potential difference of 1 V.

Q.62

EARTHING

In pipe earthing, the diameter of GI pipe embedded in the pit is

- a) 32 mm
- b) 38 mm
- c) 48 mm
- d) 56 mm

✓ ANSWER: (B)

Pipe electrodes shall not be smaller than **38 mm** internal diameter if made of galvanized iron or steel, and 100 mm internal diameter if made of cast iron.

If a resistor of 100 ohms is connected in series with a parallel combination of two 200 ohms resistors, then the effective resistance is

- a) 200 ohms b) 250 ohms
c) 350 ohms d) 150 ohms

✓ ANSWER: (A)

Series resistor: $R_1 = 100 \, \Omega$
Parallel: $R_2 = 200 \, \Omega$, $R_3 = 200 \, \Omega$
 $R_p = (200 \times 200) / (200 + 200) = 40000 / 400 = 100 \, \Omega$
Effective = $R_p + R_1 = 100 + 100 = 200 \, \Omega$

If a resistor of $20 \, \Omega$ is connected across a source of 5 volts DC supply, then the current in the circuit is

- a) 1 Ampere b) 4 Amperes
c) 0.5 Amperes d) 0.25 Amperes

✓ ANSWER: (D)

$R = 20 \, \Omega$, $V = 5 \, V$
 $I = V/R = 5/20 = 0.25 \, \text{Amperes}$

Power factor is

- a) ratio of resistance to inductance
- b) ratio of apparent power to true power
- c) ratio of resistance to impedance**
- d) ratio of inductance to capacitance

✓ ANSWER: (C)

In AC circuits, the **power factor** is defined as the ratio of resistance (R) to impedance (Z).

$$\cos \phi = R/Z$$

The phase-to-neutral voltage in a 3-phase star connected system is 230 V. The phase-to-phase (line-to-line) voltage is

- a) 230 V
- b) 398.37 V**
- c) 400 V
- d) 440 V

✓ ANSWER: (B)

$$V_L = \sqrt{3} \times V_{ph} = \sqrt{3} \times 230 = 398.37 \text{ V}$$

The time period of an AC wave at frequency of 50 Hz is

- a) 2 milliseconds
- b) 10 milliseconds
- c) 20 milliseconds**
- d) 50 milliseconds

✓ ANSWER: (C)

$$T = 1/f = 1/50 = 0.02 \text{ s} = 20 \text{ milliseconds}$$

The type of fuse used for domestic purpose is

- a) HRC fuse
- b) Kitkat or rewirable fuse
- c) Ceramic cartridge fuse
- d) Glass cartridge fuse

✓ ANSWER: (B)

The **Kitkat or rewirable fuse** is used for domestic purposes. It safeguards the circuit and appliances from damage due to voltage fluctuations and short circuits.

MCCB stands for

- a) Molded Case Circuit Breaker
- b) Miniature Case Circuit Breaker
- c) Maximum Current Circuit Breaker
- d) Minimum Current Circuit Breaker

✓ ANSWER: (A)

MCCB stands for **Molded Case Circuit Breaker**.

ELCB is used for detecting current leakage

- a) above 8 kVA
- b) below 5 kVA
- c) above 5 kVA
- d) below 8 kVA

GRACE MARKS

ELCB (Earth Leakage Circuit Breaker) is used for detecting current leakage to earth.

Q.71**MACHINES**

A static machine which transfers electrical power from one circuit to another circuit without changing the frequency is called

- a) DC machine
- b) Alternator
- c) Induction motor
- d) Transformer

✓ ANSWER: (D)

A **Transformer** is designed to either increase or decrease AC voltage between circuits while maintaining the same frequency.

Q.72**MOTORS**

The initial type of connection of the motor windings when started with a star delta starter is

- a) star connection
- b) delta connection
- c) series
- d) parallel

✓ ANSWER: (A)

During starting, the stator winding is **star connected**, which provides less starting torque and puts less mechanical and electrical stress on the system.

Q.73**MOTORS**

The cause for a 3-phase motor producing inadvertent mechanical noise is:

- a) Interchanged supply terminals
- b) High load on motor
- c) High voltage on motor winding
- d) Incorrect coupling

✓ ANSWER: (D)

If the coupling is not aligned perfectly, it can cause excessive wear and eventually lead to failure.

Incorrect coupling is a primary cause of mechanical noise.

Cell is an _____ device

- a) electro-mechanical
- b) electro-chemical**
- c) electro-magnetic
- d) electro-dynamic

✓ ANSWER: (B)

An **electrochemical** cell generates electrical energy from chemical reactions or uses electrical energy to facilitate chemical reactions.

The most commonly used battery in electric vehicles is

- a) Lithium-ion battery**
- b) Lead-acid battery
- c) Nickel-Cadmium battery
- d) Alkaline rechargeable battery

✓ ANSWER: (A)

Lithium-ion batteries are popular due to their light weight, high energy density, low self-discharge, no memory effect, and ability to recharge.

Digital signals are characterized by

- a) Continuous voltage levels
- b) Infinite resolution
- c) Discrete voltage levels**
- d) Variable voltage levels

✓ ANSWER: (C)

A digital signal can only take on one value from a finite set of possible values at a given time — **discrete voltage levels**.

Q.77

BOOLEAN ALGEBRA

According to rules and laws of Boolean Algebra, $A + A =$

- a) $2A$ **b) A**
- c) 1 d) A^2

✓ ANSWER: (B)

By the **Idempotent Law** of Boolean Algebra: $A + A = A$ and $A \cdot A = A$

Q.78

ELECTRONICS

Photo diode is used in which of the following applications?

- a) Voltage regulation b) Temperature measurement
- c) Light detection** d) Radio Frequency (RF) Amplification

✓ ANSWER: (C)

Photodiodes generate current flow directly depending on light intensity, making them suitable for **light detection** and optical signal detection.

Q.79

RESISTORS

If a resistor is having a colour band as first band = Brown, second band = Black and third band = Red, then the resistance value is

- a) $1 \text{ k}\Omega$** b) $10 \text{ k}\Omega$
- c) 100Ω d) $100 \text{ k}\Omega$

✓ ANSWER: (A)

1st band (Brown) = 1
2nd band (Black) = 0
3rd band (Red) = multiplier $\times 100$
Resistance = $10 \times 100 = 1000 \Omega = 1 \text{ k}\Omega$

The binary equivalent of the decimal number 9 is

a) 1001

b) 1000

c) 1100

d) 1010

✓ ANSWER: (A)

$9 \div 2 = 4$ remainder 1

$4 \div 2 = 2$ remainder 0

$2 \div 2 = 1$ remainder 0

$1 \div 2 = 0$ remainder 1

$\therefore (9)_{10} = (1001)_2$

05

CHAPTER FIVE

Project Management Skills

20 Marks

Questions 81 – 100

Q.81

PM BASICS

Project Management is a combination of

- a) human and non-human resources
- b) only human resources
- c) only non-human resources
- d) no resources at all

✓ ANSWER: (A)

Project management involves the coordination and utilization of both **human resources** (managers, team members, stakeholders) and **non-human resources** (tools, equipment, technology, materials, finances).

Q.82

CONSULTANTS

The consultant who is appointed to carry out the project work is

- a) Compound house consultant
- b) In-house consultant
- c) Out-house consultant
- d) Bridge consultant

✓ ANSWER: (C)

An **outsourced (out-house) consultant** is someone hired from outside the organization to provide expertise, guidance, and support for a specific project.

Q.83

PROJECT TYPES

The type of project which requires minimum amount of capital is

- a) Crash project
- b) Normal project
- c) Disaster project
- d) Consultant project

✓ ANSWER: (B)

Normal projects are allowed to take their normal time, with minimum capital and no sacrifice in quality.

Projects like building a hospital, a park, a playground and government projects like highway construction are examples of

- a) Social needs
- b) Customer needs
- c) Market needs
- d) Ecological needs

✓ ANSWER: (A)

Hospitals, parks, schools, and road construction are basic requirements for society — hence they are **social need** projects.

In Project Management, WBS stands for

- a) Work Breakdown Structure
- b) Waste Breakdown Structure
- c) Window Breakdown Structure
- d) Wireless Breakdown Structure

✓ ANSWER: (A)

WBS stands for **Work Breakdown Structure**.

The first step in Project Execution Plan is

- a) Work packaging plan
- b) Contracting plan
- c) Organization plan
- d) Procedure plan

✓ ANSWER: (B)

Under the four sub-plans of PEP, the **contracting plan** is the first step. It is necessary to contract out areas where the owner company does not have inherent competence.

Q.87

TEAMS

The team which gives the idea to start a project is

- a) Core project team
- b) Full project team
- c) Advising project team
- d) Initial project team

✓ ANSWER: (D)

The **initial project team** consists of specific people who initially conceive the idea of starting a project.

Q.88

ABBREVIATIONS

In PEP, the letter 'E' stands for

- a) Execution
- b) Estimation
- c) Evaluation
- d) Enthusiasm

✓ ANSWER: (A)

PEP stands for Project **Execution** Plan.

Q.89

PROJECT LIFE CYCLE

In Project Life Cycle, more time is required for

- a) Project closure
- b) Project initiation
- c) Project execution
- d) Project planning

✓ ANSWER: (C)

The **execution phase** consumes maximum time to complete work in a project life cycle.

Innovation is the hallmark of every project. Here innovation means

- a) New ideas
- b) Project success
- c) Professional approach
- d) Project Management

✓ ANSWER: (A)

Innovation refers to the process of introducing **new ideas**, methods, products, services, or approaches that create value and improve upon existing practices.

The Project Life Cycle Curve indicates

- a) Work packaging
- b) Number of workers in the project
- c) Growth, maturity and decline
- d) Project manual

✓ ANSWER: (C)

The project life cycle curve explains how a project is initiated, its **growth, maturity, and decline** (completion).

In Project Management, taking actions to measure the quality accurately is the function of

- a) Quality management
- b) Cost management
- c) Review management
- d) Risk management

✓ ANSWER: (A)

Quality Management is the method to measure quality accurately in Project Management.

Project planning is

- a) Planning by obligation
- b) Planning by incentive and direction**
- c) Planned initiation
- d) Changing the project policies

✓ ANSWER: (B)

Project planning involves creating incentives to motivate team members and providing clear guidance and **direction** to the project team.

Identify the incorrect statement below:

- a) Project objective should be specific
- b) Project objective should be realistic
- c) Project objective should not be framed timely**
- d) Project objective should be measurable

✓ ANSWER: (C)

Project objectives must be specific, measurable, and realistic. They should be framed within a timeline — so the statement "should not be framed timely" is **incorrect**.

WBS, PEP and PPM are the tools used to design

- a) Project plan
- b) Project work system**
- c) Project diary
- d) Project direction

✓ ANSWER: (B)

Project Work System classifies into Work Breakdown Structure, Project Execution Plan, and Project Procedure Manual.

The earliest method used for planning of project was

- a) CPM
- b) PERT
- c) Bar Chart**
- d) Milestone Chart

✓ ANSWER: (C)

The earliest method used for project planning is the **Bar Chart** (also known as Gantt Chart).

The expansion of PERT is

- a) Programme Evaluation and Review Technique**
- b) Project Estimation and Review Technique
- c) Project Estimation and Resource Technology
- d) Performance Estimation and Resource Tool

✓ ANSWER: (A)

PERT stands for **Programme Evaluation and Review Technique**.

For non-repetitive projects _____ tool is used in production planning and scheduling.

- a) CPM
- b) PERT**
- c) Both CPM and PERT
- d) Bar Chart

✓ ANSWER: (B)

For non-repetitive projects, **PERT** is the tool used in project planning and scheduling.

Q.99

REVIEW

The purpose of conducting a project review is

- a) To finance the project
- b) To initiate the project
- c) To develop the project scope
- d) To assess project performance

✓ ANSWER: (D)

Project review helps in evaluating and **assessing project performance**.

Q.100

REVIEW

A project review does not contain

- a) Performance evaluation
- b) Evaluating the capital budget
- c) Data collection
- d) Initial review

✓ ANSWER: (C)

Data collection is not a step of project review.



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